THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 12: **Tales by Dots and Lines** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 42	$5 \times 30 - 4 \times 27$
2	**(d)** 41	Correct sum $= 360 - 30 + 80$
3	**(c)** 43	$(29 + 1)(14) - 29 \times 13$
4	**(a)** 12	Sum of first n odd numbers is n^2, so mean $= n$
5	**(c)** 40	Sort the 8 values; median is the mean of the 4th and 5th: 40
6	**(b)** 48	Total distance 240 over total time
7	**(d)** 49	Mean is multiplied by k, then c is added
8	**(a)** 36	$5 \times 24 - 4 \times 21$
9	**(a)** Both A and R are true and R is the correct explanation of A.	$\frac{4+6+8+10+12}{5} = \frac{40}{5} = 8$; R is the definition used.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	The middle value of the five ordered numbers is 9, so A is true; R is also true, but mode is a different measure and does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Total $= 6 \times 20 = 120$ (1); rest $= 5 \times 18 = 90$, removed $= 30$ (1).	1+1
12	Sum $= 2 + 4 + \cdots + 16 = 72$ (1); mean $= 72 \div 8 = 9$ (1).	1+1
13	Wrong sum $= 198$ (1); correct sum $= 192$; mean $= 32$ (1).	1+1
14	Total $= 6 \times 35 = 210$ (1); rest $= 5 \times 32 = 160$, removed $= 50$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	New sum $= 2480 - 38 + 80 = 2522$ (2); mean $= 63.05$ (1).	2+1
16	Total age of players $= 253$ (1); total with coach $= 300$ (1); coach $= 47$ years (1).	1+1+1
17	Required total $= 6 \times 68 = 408$ (1); present total $= 310$ (1); needed $= 98$ (1).	1+1+1
	OR: New sum $= 1200 - 20 + 70 = 1250$ (2); mean $= 50$ (1).	2+1
18	Total age of players $= 300$ (1); total with coach $= 352$ (1); coach $= 52$ years (1).	1+1+1
	OR: Required total $= 6 \times 68 = 408$ (1); present total $= 310$ (1); needed $= 98$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{98}{7} = 14$ (1). (ii) Ordered: 9, 10, 12, 15, 15, 18, 19; median $= 15$ (1). (iii) Mode $= 15$ (1). (iv) Range $= 19 - 9 = 10$ (1); new mean $= \frac{98 + 20}{8} = 14.75$ (1).	1+1+1+2
20	Total $= 10 \times 40 + 20 \times 50 + 20 \times 30 = 2000$ (3); mean $= 40$ (2).	3+2
	OR: New mean $= 20 \times 3 - 8 = 52$ (3); new sum $= 15 \times 52 = 780$ (2).	3+2

Mistakes to avoid:

- Finding the median without arranging the data in order first.
- For an even number of observations, taking one of the middle values instead of the **mean of the two middle values**.
- Confusing mode with median or mean: mode is the most frequent value, not the middle value.
- Dividing the sum by the number of *distinct* values instead of the total number of observations (counting repeated values only once).
- Computing range as the difference between the first and last values as listed, not the maximum and minimum.
- Reading a line graph at the wrong point or ignoring the scale of the axes.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 12: **Tales by Dots and Lines** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 15	Sum of first n odd numbers is n^2, so mean $= n$
2	**(b)** 46	Sort the 8 values; median is the mean of the 4th and 5th: 46
3	**(b)** 36	Total distance 180 over total time
4	**(d)** 51	Mean is multiplied by k, then c is added
5	**(c)** 65	$5 \times 45 - 4 \times 40$
6	**(d)** 62	Correct sum $= 900 - 45 + 75$
7	**(d)** 49	$(34+1)(15) - 34 \times 14$
8	**(c)** 21	Sum of first n odd numbers is n^2, so mean $= n$
9	**(c)** A is true but R is false.	5 occurs most often, so A is true. R is false: the middle value is the median.
10	**(d)** A is false but R is true.	Median $= \frac{5+8}{2} = 6.5$, so A is false; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sum $= 2 + 4 + \cdots + 22 = 132$ (1); mean $= 132 \div 11 = 12$ (1).	1+1
12	Wrong sum $= 216$ (1); correct sum $= 210$; mean $= 35$ (1).	1+1
13	Total $= 6 \times 14 = 84$ (1); rest $= 5 \times 12 = 60$, removed $= 24$ (1).	1+1
14	Sum $= 2 + 4 + \cdots + 28 = 210$ (1); mean $= 210 \div 14 = 15$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	New sum $= 1650 - 36 + 66 = 1680$ (2); mean $= 56$ (1).	2+1
16	Total age of players $= 350$ (1); total with coach $= 405$ (1); coach $= 55$ years (1).	1+1+1
17	Required total $= 6 \times 67 = 402$ (1); present total $= 305$ (1); needed $= 97$ (1).	1+1+1
	OR: New sum $= 1440 - 52 + 92 = 1480$ (2); mean $= 74$ (1).	2+1
18	Total age of players $= 270$ (1); total with coach $= 320$ (1); coach $= 50$ years (1).	1+1+1
	OR: Required total $= 6 \times 68 = 408$ (1); present total $= 310$ (1); needed $= 98$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $2 + 4 + 3 + 1 = 10$ (1). (ii) Mode $= 1$ (1). (iii) $3 + 1 = 4$ students (1). (iv) Mean $= \frac{0+4+6+3}{10} = 1.3$ (1); ordered data has 5th and 6th values both 1, so median $= 1$ (1).	1+1+1+2
20	Total $= 20 \times 30 + 10 \times 25 + 20 \times 70 = 2250$ (3); mean $= 45$ (2).	3+2
	OR: New mean $= 30 \times 2 - 11 = 49$ (3); new sum $= 10 \times 49 = 490$ (2).	3+2

Mistakes to avoid:

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- Dividing the sum by the number of *distinct* values instead of the total number of observations (counting repeated values only once).
- Computing range as the difference between the first and last values as listed, not the maximum and minimum.
- Reading a line graph at the wrong point or ignoring the scale of the axes.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 12: **Tales by Dots and Lines** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 73	Mean is multiplied by k, then c is added
2	**(a)** 10	$5 \times 18 - 4 \times 20$
3	**(b)** 53	Correct sum $= 600 - 26 + 62$
4	**(b)** 31	$(19+1)(12) - 19 \times 11$
5	**(d)** 30	Sum of first n odd numbers is n^2, so mean $= n$
6	**(b)** 30.5	Sort the 8 values; median is the mean of the 4th and 5th: 30.5
7	**(d)** 60	Total distance 300 over total time
8	**(a)** 46	Mean is multiplied by k, then c is added
9	**(a)** Both A and R are true and R is the correct explanation of A.	$40 - 7 = 33$; R is the definition.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	A dot plot puts one dot for each occurrence of a value, so A is true; R is also true but does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Wrong sum $= 198$ (1); correct sum $= 192$; mean $= 32$ (1).	1+1
12	Total $= 6 \times 48 = 288$ (1); rest $= 5 \times 45 = 225$, removed $= 63$ (1).	1+1
13	Sum $= 2 + 4 + \cdots + 40 = 420$ (1); mean $= 420 \div 20 = 21$ (1).	1+1
14	Wrong sum $= 168$ (1); correct sum $= 162$; mean $= 27$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	New sum $= 2050 - 30 + 60 = 2080$ (2); mean $= 41.6$ (1).	2+1
16	Total age of players $= 323$ (1); total with coach $= 380$ (1); coach $= 57$ years (1).	1+1+1
17	Required total $= 6 \times 60 = 360$ (1); present total $= 270$ (1); needed $= 90$ (1).	1+1+1
	OR: New sum $= 1540 - 35 + 70 = 1575$ (2); mean $= 45$ (1).	2+1
18	Total age of players $= 672$ (1); total with coach $= 750$ (1); coach $= 78$ years (1).	1+1+1
	OR: Required total $= 6 \times 63 = 378$ (1); present total $= 285$ (1); needed $= 93$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Wednesday, $33°$C (1). (ii) $33 - 28 = 5°$C (1). (iii) Thursday to Friday, a fall of $3°$C (Wednesday to Thursday fell $2°$C) (1). (iv) Mean $= \frac{28+30+33+31+28}{5} = \frac{150}{5} = 30°$C (1); days above the mean: Wednesday and Thursday, 2 days (1).	1+1+1+2
20	Total $= 10 \times 35 + 20 \times 15 + 20 \times 70 = 2050$ (3); mean $= 41$ (2).	3+2
	OR: New mean $= 16 \times 3 - 7 = 41$ (3); new sum $= 25 \times 41 = 1025$ (2).	3+2

Mistakes to avoid:

- Finding the median without arranging the data in order first.
- For an even number of observations, taking one of the middle values instead of the **mean of the two middle values**.
- Confusing mode with median or mean: mode is the most frequent value, not the middle value.
- Dividing the sum by the number of *distinct* values instead of the total number of observations (counting repeated values only once).
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